

A Textual-Visual-Entailment-based Unsupervised Algorithm for Cheapfake Detection

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ABSTRACT

The growth of communication has led to misinformation in many different forms. "Cheapfake" is a recently coined term referring to manipulated media generated by non-AI techniques. One of the most prevalent ways to create cheapfakes is by simply altering the context of an image/video using a misleading caption. The "ACMMM 2022 Grand Challenge on Detecting Cheapfakes" has raised the problem of catching the out-of-context misuse to assist fact-checkers, as detecting conflicting image-caption sets helps narrow the search space. To cope with this challenge, we propose a multimodal heuristic method. The proposed method expands the baseline method of the challenge (i.e., COSMOS) with four additional components (i.e., Natural Language Inference, Fabricated Claims Detection, Visual Entailment, and Online Caption Checking) towards overcoming the current weaknesses of the baseline. During runtime, our proposed method has achieved a maximum of 89.1% accuracy on Task 1, which is 7.2% higher than the baseline method, and 73% accuracy on Task 2. The code for our solution is publicly available on Github ¹ and the Docker image can be found on DockerHub ².

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**;
Computer vision.

¹<https://github.com/pwnyniche/acmmmcheapfake2022>

²<https://hub.docker.com/repository/docker/tqtnk2000/acmmmcheapfakes>

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KEYWORDS

Semantic Textual Similarity, Computer Vision, Natural Language Inference, Cheapfakes, Misinformation, Visual Entailment, Online Caption Checking, Fabricated Claims Detection

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1 INTRODUCTION

Although we are no doubt of the benefits of the Internet, it has its dark side. Altering the context of an image/video and sharing the media alongside misleading claims is one of the dark side problems, namely cheapfakes. Cheapfakes can be created without using software by simply altering the context of an image/video by sharing the media alongside misleading claims. This alteration of context is referred to as out-of-context (OOC) misuse of media.

To cope with this problem, "ACMMM 2022 Grand Challenge on Detecting Cheapfakes" [2] has raised two challenges Task 1 and Task 2. The former is to detect conflicting image-caption triplets $\langle \text{image}, \text{caption1}, \text{caption2} \rangle$, which indicates miscontextualization. If the captions refer to the same object in a given image but imply semantically different meanings, they are considered Out-of-Context (OOC). On the contrary, the captions that semantically resemble or target different objects are classified as Not-Out-of-Context (NOOC). The latter is to determine whether a given $\langle \text{Image}, \text{Caption} \rangle$ pair is genuine (real) or falsely generated (fake). This task focuses on validating the veracity of a statement given an image.

2 METHOD

This section introduces the heuristic multimodal method that leverages textual and visual entailment for cheapfake detection. We inherit the baseline algorithm of this challenge (i.e., COSMOS) as the fundamental component. Then, we enrich the baseline by integrating Natural Language Inference (NLI), Fabricated claims

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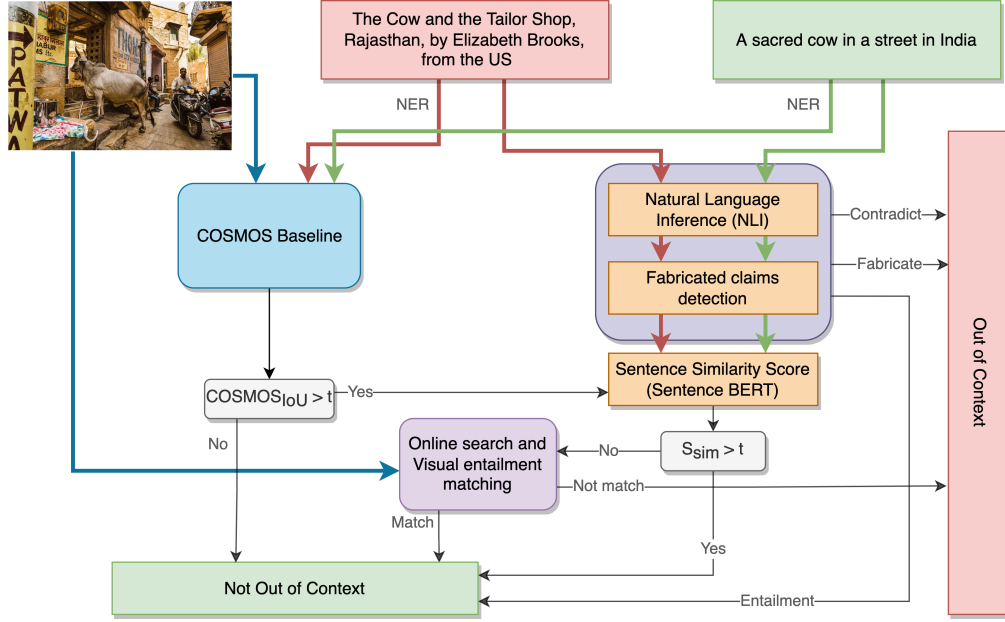


Figure 1: Method's Overview. First, the two captions are fed through the NLI model to detect Contradiction and Entailment and labeled as OOC and NOOC, respectively. Results assigned as Neutral then go to the FCD component. The output of FCD component is labeled as OOC if it is fabricated. The rest of the inputs are put through the COSMOS baseline, and those captions that overlap the same region on the image ($COSMOS_{IoU} > t$) but have a low similarity score ($S_{sim} \leq t$) are boost with our Online search and Visual entailment component.

detection (FCD), Visual Entailment (VE), and Online Caption Checking (OCC) components to gain better accuracy. The whole method is intended to solve the first task of the challenge, and some of the work is reprocessed to handle the second task. Figure 1 illustrates the overview of the proposed method.

2.1 Method's Overview

The proposed method leverages the if-else manner (i.e., heuristic approach) to switch the detection flow into different cases defined after reasonably observing several cases that failed using the baseline, as described in Figure 1.

First, we recognize that the baseline model almost failed when one caption is copied from the same source with an image but has no relation that can extract directly from them, and another is the image's description. Another case is that even though both captions are from the same source with the image (i.e., they might talk about the same topic), their sentence similarity score is low if using a textual model for comparing. To overcome this obstacle, we propose the component of online search and visual-text matching (i.e., visual entailment).

Second, we observe that a large portion of caption pairs can be concluded as OOC or NOOC by using Natural Language Inference (NLI) model, with the relative result being Contradiction or Entailment, respectively. Unfortunately, if the result is Neutral, we cannot get a conclusion. Hence, the component of NLI is integrated into our method.

Third, we build the Fabricated claim detection component to overcome the Neutral case mentioned above.

2.2 Component Details

2.2.1 Natural Language Inference. Natural Language Inference (NLI) which is also known as Recognizing Textual Entailment (RTE), is a task of determining whether the given "hypothesis" and "premise" logically follow (entailment) or unfollow (contradiction) or are undetermined (neutral) with each other.

The COSMOS method calculates the OOC/NOOC by leveraging the semantic textual similarity (STS) score. Unfortunately, there are many cases where the STS score does not perform well. For example, even though the two captions are contradictory, the STS model may produce a high score if the ratio of similar words is high. In contrast, a pair of entailment captions may have a low STS score if one caption is much more detailed than the other. To overcome this obstacle, we utilize the NLI model to boost the result in cases the prior method could not handle.

We examined several SOTA models on the SNLI and MNLI datasets [3, 9]. Then, we choose the DeBERTa model [5] on MNLI as the best-performing model for the challenge's dataset. The NLI model takes two captions: C_1 as the premise, C_2 as the hypothesis, and returns a label $\mathcal{L}_{nli}$ that can either be Contradiction, Entailment, or Neutral, along with the confidence score S_{nli} . Since the two captions should share the same impact, we run the model twice by switching the premise and the hypothesis, as described in Equations (1) and (2).

$$\mathcal{L}_1 = \mathcal{L}_{nli}(C_1, C_2) \quad (1)$$

as described in Equation (7).

$$S_{CLIP} = CLIP(I_1, I_2) \quad (7)$$

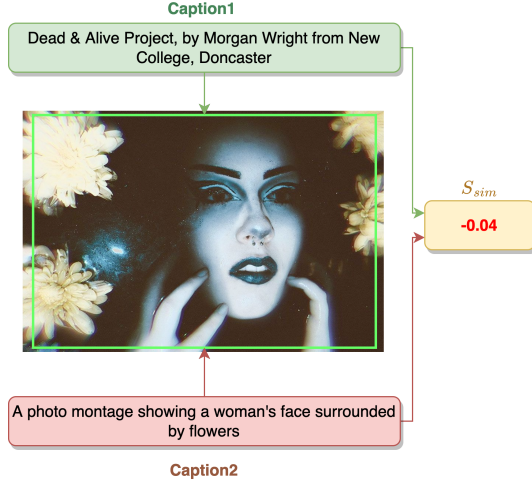


Figure 3: Illustration of the case that two captions overlap the image region with low SBERT score, but still not out-of-context.

Suppose the similarity scores S_{CLIP} of the caption C is higher than the threshold $t_c = 0.97$, we conclude the caption and the image are in the same context. Integrating with the VE model mentioned above, we label the triplets NOOC according to Equation (8).

$$\mathcal{L}(C_1, C_2) = NOOC, \quad \text{if } \begin{cases} S_{CLIP}(C_i) > t_c \\ S_{VE}(C_i) > t_v \\ S_{CLIP}(C_i) > t_c \ \& \ S_{VE}(C_{j \neq i}) > t_v \end{cases} \quad i, j \in \{1, 2\} \quad (8)$$

2.3 Task 2

We leverage the online search and visual entailment components developed for Task 1 to check the veracity of an image-caption pair. We predict the label of the image-caption pair using Equation (9).

$$\mathcal{L}(C) = NOOC, \quad \text{if } \begin{cases} S_{CLIP}(C) > t_c \\ S_{VE}(C) > t_v \end{cases} \quad (9)$$

For task 2, the test dataset lacks sentences describing the image's content. Therefore, the S_{VE} entailment score is low for all the image-caption pairs, and it is impossible to predict any entailments using the current threshold $t_v = 0.3$. However, we expect the VE model to be more helpful on a more extensive and varied test dataset.

3 EXPERIMENTS AND RESULTS

The challenge provides a large-scale dataset containing 200K images and 450K corresponding captions in which some images may have more than one caption. The photos are gathered from various news websites, fact-checking websites, and social media posts. For benchmarking purposes, the challenge manually annotated a subset of 1700 triplet pairs (i.e., one picture and two captions) to evaluate the baseline model. In the end, the baseline technique produces an OOC detection accuracy of over 85%. For the second task, the

authors extracted 100 images from the test dataset, each image will have a caption associated with a genuine/fake label.

We use the pre-trained SBERT-WK model in the COSMOS baseline to calculate the semantic textual similarity score for all of our methods. For the NLI method, we use the *deberta-xlarge-mnli* model to output the Contradiction, Entailment, and Neutral scores. For the visual entailment method, we use the finetuned OFA-large model[7] on SNLI-VE[10] dataset. We only use the image premise and do not use the text premise as the input. We follow Akgul et al. [1] and reduce the COSMOS's IoU threshold to 0.25 for all of our experiments. Table 1 summarizes the results of our methods for the given test dataset.

Model	Accuracy	Precision	Recall	F1
Baseline	0.819	0.852	0.772	0.810
Baseline + NLI	0.871	0.870	0.872	0.871
Baseline + FCD	0.860	0.838	0.892	0.864
Baseline + OCC + VE	0.849	0.885	0.802	0.842
All methods	0.891	0.878	0.908	0.893

Table 1: Results for the test dataset.

For Task 2, we reuse our online searching and VE method, keeping all the configurations the same and achieving the accuracy of 73%.

4 CONCLUSIONS AND DISCUSSIONS

This paper presents a multimodal heuristic method to detect the OOC misuse of media. We address the relation between two captions using the Natural Language Inference task and a strategy to detect misleading captions. We also propose an online "cheapfake" detector that can be useful to capture the caption's honesty and to check the veracity of an image-caption pair. We achieved 7.2% accuracy over the COSMOS baseline for Task 1 and attained the accuracy of 73% for Task 2.

The main idea of online caption checking is to see whether captions and an image appear/entail in the same article. We hypothesize that there must be an online image representing an event, then people want to alter the context. Next step, we will query by images and find all related articles. Then, we will check the entailment of captions in these articles. Besides, we will develop a new model to extract the named entities (e.g., places, times, celebrities' names) in the sentence and image and then utilize a knowledge graph to examine the relationships between these entities to ensure they are in the same context.

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